

Lecture 1

◆ Definition

The **Internet of Things (IoT)** is a network of smart physical objects (devices, vehicles, buildings, etc.) that have sensors, computing power, memory, energy source, and network connectivity.

These devices can **collect, exchange, and analyze data** to make intelligent decisions.

Goal: “Connect the unconnected.”

◆ Main Technologies

IoT combines several technologies:

- Embedded Systems
 - Low-Power Networks
 - Cloud Computing
 - Internet
 - Data Analytics
 - Big Data & Edge Intelligence
 - Network & Data Security
 - Software-Defined Networks
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◆ Other Names of IoT

- Internet of Everything
 - Smarter Planet
 - Machine-to-Machine (M2M)
 - The Fog
 - Tsensors (Trillion Sensors)
 - The Industrial Internet / Industry 4.0
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◆ Benefits of IoT

1. **Real-Time Monitoring** – Continuous observation of devices and processes.
2. **Automation** – Machines perform tasks faster and with fewer errors.
3. **Improved Efficiency** – Streamlines operations and boosts productivity.
4. **Predictive Maintenance** – Detects problems before failure occurs.
5. **Better Insights** – Data analysis reveals trends and areas to improve.
6. **Cost Reduction** – Less downtime, fewer errors, and optimized resources.

7. **Optimized Work Environments** – More comfortable and efficient spaces.
 8. **Adaptability** – Easily adjusts to new business needs or growth.
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◆ Main Challenges

1. **Security:** Many connected devices make protection complex.
2. **Privacy:** Deciding what personal data to share and how to control it.
3. **Low-Power Networks:** Devices must stay connected for years using minimal energy.
4. **Interoperability:** Different protocols and platforms cause compatibility issues.
5. **Sensor Scale:** Limited sensors and resources in some environments.
6. **Big Data & Analytics:** Handling and analyzing massive, diverse data efficiently.

Lecture 2

🧠 1. IoT Communication

Anytime Communication:

Communication can happen at any time — day or night, indoors or outdoors, at the computer or on the move.

(يعني التواصل في أي وقت)

Anything Communication:

Communication types include:

- Human ↔ Human (إنسان لإنسان)
- Human ↔ Thing (إنسان لجهاز)
- Thing ↔ Thing (جهاز لجهاز)
- Computer ↔ Computer

⚙️ 2. Building Blocks of IoT (مكونات إنترنت الأشياء الأساسية)

1. **Sensors/Devices (المستشعرات أو الأجهزة)**
 - Collect real-time data from the environment.
 - Have unique IP addresses.
 - Work automatically or manually.
 - Examples: Gas sensor, Moisture sensor, Water quality sensor.
2. **Processors (المعالجات / وحدات التحكم)**
 - The “brain” of IoT — processes and analyzes data.
 - Performs data encryption and decryption (تشفير البيانات لحمايتها).
 - Examples: Microcontrollers, Embedded systems.
3. **Gateways (البوابات)**

- Connect IoT devices to the network or cloud.
- Manage data transfer and communication between systems.
- Examples: Routers, IoT Hubs.

4. Applications (التطبيقات)

- The user-facing part of IoT that uses data for smart actions.
- Examples: Smart home apps, Security systems, Industrial dashboards.

3. Building an IoT Architecture (بناء هيكل إنترنت الأشياء)

When designing an IoT architecture, you must define the **problem domain**, **technologies**, and **system solution**.

Step 1: Define the Problem Domain

Start by understanding the needs, problems, and constraints of your system.

- Application needs and scenarios (احتياجات وسيناريوهات التطبيق)
- Requirements (المتطلبات)
- Business and technical constraints (القيود التجارية والتقنية)

Representation:

/ Needs \
| Problems |
\ Constraints /

Step 2: Define Concepts and Base Technologies

Turn the problem into a possible solution using:

- Concepts (المفاهيم)
- Base technologies (التقنيات الأساسية)

For this step, define:

- Design objectives and principles (أهداف ومبادئ التصميم)
- Architecture views (وجهات النظر المعمارية)
- Needed capabilities and functions (القدرات والوظائف المطلوبة)
- Technology components (العناصر التقنية)

Step 3: System Solutions

Finally, create the full system design:

- System design (تصميم النظام)
- Deployment view (طريقة نشر النظام)
- Software components (مكونات البرمجيات)

- Hardware components (مكونات الأجهزة)

4. IoT Design Principles (مبادئ تصميم أنظمة إنترنت الأشياء)

1. Reuse resources (إعادة استخدام الموارد)
2. Open services and APIs (خدمات مفتوحة للتكامل)
3. Design for sensing and actors taking on different roles of providing and using services across different business domains and value chains. (تعدد المجالات)
4. Design for ensuring trust, security, and privacy.
5. For Scalability and performance (قابلية التوسع والأداء العالي)
6. For the Evolution and integration (سهولة التطوير والتكامل)
7. For the Management simplification (تبسيط الإدارة)
8. With Cloud-based design (تصميم يعتمد على السحابة)
9. For Full lifecycle support (دعم النظام في كل مراحله)

5. IoT Architectural Layers (طبقات هيكل إنترنت الأشياء)

There are multiple layers that work together to form an IoT system:

Layer	Description	Arabic Translation
1. Asset Layer	This layer is, strictly speaking, not providing any functionality within a target solution, but justification for existence for any IoT application	الطبقة الفيزيائية للأجهزة
2. Resource Layer	Sensors, actuators, and smart devices.	طبقة الموارد
3. Communication Layer	Provides connectivity (LAN, WAN, Internet).	طبقة الاتصال
4. Service Support Layer	Offers support services (cloud, servers).	طبقة دعم الخدمات
5. Data & Information Layer	Converts data into knowledge through analytics.	طبقة البيانات والمعلومات
6. Application Layer	Real IoT apps like Smart Grid or Smart Home.	طبقة التطبيقات
7. Business Layer	supporting the core business or operations of any enterprise, organization, or individual that is interested in IoT applications. Customer Relationship Management (CRM), Enterprise Resource Planning (ERP), or other Business Support Systems (BSS).	طبقة الأعمال والإدارة

6. Cross-Layer Functional Groups (وظائف تعمل عبر جميع الطبقات)

1. **Management** → Managing the operation, maintenance, and performance of devices and services.
 2. **Security** → Protecting the system and data from external threats (communication and information security).
 3. **IoT Data & Services** → Handling data flow, event filtering, aggregation, and transforming data into knowledge using analytics.
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Lecture 3

IoT Architecture Models

1. Basic 3-Layer Architecture

1. **Perception Layer**
 - Physical layer with sensors gathering environmental data.
2. **Network Layer**
 - Connects devices and transfers data to servers and other systems.
3. **Application Layer**
 - Delivers application-specific services such as smart homes, cities, and healthcare.

2. Common 4-Layer Architecture

1. **Sensing Layer**
 - Sensors and actuators collect data from the environment.
 2. **Network Layer**
 - Protocols and technologies for device connectivity and internet access.
 3. **Data Processing Layer**
 - Data management, analytics, and machine learning for decision-making.
 4. **Application Layer**
 - Software interfaces like web portals and mobile apps interacting with IoT systems.
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Traditional IoT Data Flow

IoT devices collect data → send through IoT network → travel via internet → processed by systems → viewed by users/administrators.

Data Processing Locations

Processing can occur at:

- **Cloud**
- **Edge**
- **Local devices**

Edge Computing and Edge Intelligence

- Computation happens near data sources for faster decisions.
- AI processes data at the edge to extract insights in real time.

Key Knowledge Areas for IoT Development

Developers must understand:

- **IoT Components**
 - **Architecture**
 - **Network Access Technologies**
 - **Data Storage & Processing**
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IoT Ecosystem Components

1. **Cloud**
 2. **Analytics**
 3. **IoT Core (Sensors, Actuators, MCU, Security)**
 4. **IoT Gateway**
 5. **User Interface / Visualization**
 6. **IoT Architectures (3-layer, oneM2M, etc.)**
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Core Components of IoT

- **Sensors** – Sense and gather data
 - **Actuators** – Perform actions based on commands
 - **Microcontrollers** – Control sensors and actuators
 - **Internet Connectivity** – Communication medium
 - **Service Platforms** – Device & data management, analytics, security
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Classification of Sensors

- **By energy:** Active / Passive
- **By placement:** Invasive / Non-invasive
- **By contact:** Contact / No-contact
- **By mechanism:** Thermoelectric, electromechanical, optical, etc.
- **By parameter:** Position, motion, pressure, humidity, chemical, biosensors, etc.
- **By industry:** Medical, agriculture, manufacturing
- **By scale:** Absolute / Relative

Actuator Classification

- Common ways to classify actuators:

✓ Type of motion they produce

- e.g. linear, rotary, one/two/three axes

✓ Power output

- e.g. high power, low power, micro power

✓ Binary / Continuous output

- Based on number of stable-state outputs

✓ Area of application

- Specific industry or vertical where they are used

✓ Type of energy

44 • e.g. mechanical energy, electrical energy, hydraulic energy, etc

Actuators by Energy Type

- **Mechanical:** Levers, screw jacks
- **Electrical:** Diodes, transistors
- **Electromechanical:** DC motors, stepper motors
- **Electromagnetic:** Solenoids
- **Hydraulic/Pneumatic:** Cylinders, valves
- **Smart Materials:** Piezoelectric devices, bimetal strips

Access Technologies in IoT

Communication Criteria

- ➤ Range
- ➤ Frequency Bands
- ➤ Power Consumption
- ➤ Topology
- ➤ Constrained Devices
- ➤ Constrained-Node
- Networks

Key Advantages of IP

- Open and standard-based
- Versatile
- Ubiquitous
- Scalable
- Manageable
- Highly secure
- Stable and resilient

- WPAN: Wireless Personal Area Networks
- IEEE 802.15.4: Low-Rate WPAN

- 6LoWPAN: IPv6 over Low-Power WPAN
- TSCH: Time Synchronized Channel Hopping
- 6TiSCH: IPv6 over the TSCH mode of IEEE 802.15.4e
- 6top: 6TiSCH Operation Sublayer

IoT Gateway

- Provides bidirectional communication between IoT and other networks
 - Supports edge processing
 - Provides security barrier
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Cloud in IoT

Benefits include:

- Scalability
- Data mobility
- Security
- Faster deployment
- Cost efficiency

IoT Data Analytics

Challenges:

- Huge real-time data volume
- Variety of formats (XML, video, SMS)
- Unstructured data
- Variable meaning and models

Smart Objects

Characteristics:

1. Sensors/Actuators
2. Processing unit
3. Memory
4. Communication module
5. Power source

Trends:

- Smaller size
- Lower power
- Higher processing capabilities
- Standardized communications